Spacecraft Constellation Fault Simulator — Comprehensive User Guide

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Part 1 – Formal Documentation Manual

1. Overview

The *Spacecraft Constellation Fault Simulator* is a Basilisk-based simulation environment with a Tkinter GUI designed to configure, simulate, and analyze spacecraft constellations under various fault and recovery conditions.
It supports multi-satellite constellations, machine learning (ML) fault detection, Deep Reinforcement Learning (DRL) task reassignment, and visualization in 2D and 3D (Vizard).

The simulator includes nine main functional tabs, a top menu bar, and shared controls (Run Simulation and Help) for context-sensitive operation.

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2. Application Layout

Top Toolbar

- **File menu**
  - *Open Configuration* – Load a previously exported JSON configuration.
  - *Save Configuration* – Export the current setup (constellation, faults, ML, DRL, visualization).
  - *Exit* – Close the simulator safely.
- **Simulation menu**
  - *Run Simulation* – Executes Basilisk simulation using current parameters.
  - *Pause/Resume Simulation* – Temporarily halt or continue a running process.
  - *Stop Simulation* – Abort current simulation.
  - *Open Results Folder* – Opens the folder containing logs and plots.
  - *Generate Summary Report* – Creates a textual summary of the last simulation.
- **DRL menu**

- *Train DRL Model* – Launches DRL training window for PPO/DQN/TDHD.
- *Load DRL Model* – Manually load pre-trained .h5 DRL model.
- *Evaluate DRL Policy* – Test policy performance without running full simulation.

- **Help menu**
  - *View Tab Help* – Opens context-sensitive help (from rw_fault_help.py).
  - *About* – Displays version, Basilisk version (1.18.0), and credits.
  - *Documentation* – Opens local or online manual (if available).

Header Area

- Left: Swinburne University logo (swinburne.png)
- Center: Application title
- Right: Buttons
  - **Run Simulation (green)** — starts a complete run.
  - **Help (grey)** — opens help for the active tab.

Status Bar

- Displays current status: “Ready”, “Running Simulation”, “Completed”, or “Error”.
- Includes progress indicators and error logs during simulation runs.

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3. Tab-Level Documentation

3.1 Constellation Tab

Used for building, managing, and visualizing satellite clusters and orbits.

Sub-tabs:

1. **Cluster Management**
   - Fields:
     - *Cluster Name* – String identifier.
     - *Number of Satellites* – 2–6 range.
     - *Formation Type* – Leader-Follower, Line, Triangle, Diamond.

        - *Separation (km)* – Recommended ≥15 km for visibility.
        - *Select Orbit* – Choose from defined orbits.
    - Buttons:
        - *Create Cluster* – Adds new cluster with defined parameters.
        - *Clear Form* – Resets input fields.
        - *View Details* – Shows cluster composition.
        - *Modify Cluster* – Opens editable dialog for cluster updates.
        - *Delete Cluster* – Removes selected cluster.
        - *Test Communication* – Verifies leader-follower data links.
        - *View Formation* – Displays formation geometry in 2D.
        - *Export Config* – Saves cluster structure to JSON.
2. **Orbit Management**
    - Add, edit, or delete orbital configurations (altitude, inclination, RAAN, eccentricity).
    - Reusable across clusters.
3. **Individual Satellites**
    - Displays all satellites with assigned orbit, cluster, and role.
    - Buttons: *Add Satellite*, *Remove Satellite*, *Assign Orbit*, *Assign Cluster*.
4. **Communication**
    - Shows link topology among satellites.
    - Provides communication test and refresh options.
    - Displays signal health indicators.

3.2 Fault Configuration Tab

Allows definition, summary, and visualization of spacecraft faults.

Sub-tabs:

1. **Fault Configuration**
    - Dropdowns: *Select Cluster*, *Select Satellite*.
    - Radio Buttons: *Friction*, *Power Limit*, *Encoder*, *Battery*.
    - Parameters:
        - *Magnitude* (e.g., 0.0005 N·m)
        - *Wheel Number* (1–4)
        - *Fault Time* (minutes)
    - Periodic Fault Section:
        - *Enable Periodic Fault*
        - *Interval (s)*, *Magnitude*, *Wheel Number*
    - Buttons: *Apply to Satellite*, *Apply to Cluster*, *Apply to All*.
    - Real-time field updates with current status text (“No fault enabled”).
2. **Fault Summary**
    - Lists all satellites with fault info (type, magnitude, wheel, time).
    - Buttons: *Clear All*, *Refresh Summary*.
3. **Cluster Faults**
    - Displays cluster-level aggregation for visualization comparison.

3.3 Fault Detection Tab

Implements machine learning–based anomaly detection with explainability.

Sub-tabs:

1. **Overview**
    - Displays system readiness (LSTM, Isolation Forest, XAI availability).
    - Real-time statistics: detections, success rate, last detection timestamp.
2. **ML Configuration**
    - Buttons:

        - *Load ML Model* – Import .keras/.h5 file.
        - *Start Detection* – Run fault detection manually.
        - *Generate XAI* – Create LIME/SHAP visuals.
        - *Simulate Results* – Test using sample data.
        - *Reset* – Clear loaded models.
        - *View Results* – Jump to results panel.
    - Threshold settings and method toggles retained in configuration.
3. **Detection Results**
    - Post-simulation analysis summary.
    - Exports JSON of detection summary and plots.
4. **XAI Explanations**
    - Shows SHAP waterfall plots and LIME tables for transparency.
5. **GNN Analysis**
    - Extended analysis for time-series and graph models.
    - Sub-tabs for anomaly detection, causal graphs, training history, detailed results.

3.4 Task Reassignment Tab

Handles DRL-based task redistribution after fault detection.

Sub-tabs:

1. **Overview**
    - Displays system readiness, system health, number of reassignments, and last analysis.
    - Buttons: *Refresh Status*, *View Decisions*, *Simulate Results*.
2. **DRL Decisions**
    - Shows DRL decision-making process: actions, confidence, reassigned tasks.
3. **Spacecraft Status**

- Displays per-satellite status, health, faults, and load.
- Includes “Task Redistribution Summary”.

4. **Reassignment Strategy**
   - Shows performance of Even, Capability-Based, and Load-Balanced strategies.
   - Export and plot comparison tools.
5. **Analysis**
   - Generates visual analytics: system health over time, task distribution, confidence, performance metrics.
   - Exports results to “Analysis Results” table.

3.5 Targets Tab

Defines observation targets and assignments.

Controls:

- Buttons: *Add Target*, *Remove Target*, *Generate Random*, *Auto-Assign All*, *Clear All Assignments*, *Check Coverage*.
- Inputs: *Name, Latitude, Longitude, Priority (1–5), Color*.
- Assignment tools: *Assign Target*, *Unassign Target*.
- Coverage Map: dynamic Matplotlib preview.
- Visible status updates (“VISIBLE IN VIZARD”, “NOT VISIBLE”).

3.6 Visualization Tab

Manages visual aspects of simulation.

Sub-tabs:

1. **Camera & View**
   - Controls camera offset (X, Y, Z) and FOV.
   - Presets: Earth, Target, Side, Front, Close, Wide.
   - Buttons: *Apply Settings*, *Auto Setup*, *Disable All*.

2. **Orbit Control**
    - Show/hide orbit paths.
    - Width and transparency sliders.
    - Coloring: per cluster, satellite, orbit.
3. **Satellite Display**
    - Controls labels, marker size, color palette (Distinct, Warm, Cool, Cluster).
    - Faulted satellites highlighted automatically.
4. **Target Settings**
    - Adjust marker size and connection line visibility.
    - Display coverage zones and colors by priority.
5. **Visualization Preview**
    - Real-time simplified 3D preview before Vizard export.

3.7 Output Settings Tab

Handles simulation output options and directories.

Controls:

- *Simulation Time* (minutes), *Quick Presets* (10, 30, 60, 90).
- Checkboxes: *Show Plots*, *Save Plots*, *Save Binary*.
- *Binary Filename* field (auto timestamped).
- Directory paths:
    - *Logs Directory*
    - *Plots Directory*
    - *Vizard Files Directory*
- Buttons: *Browse*, *Reset to Defaults*.
- Simulation Log window with scroll, auto-refresh, filtering (INFO/WARN/ERROR).

3.8 Results Tab

Displays generated plots and telemetry outputs.

Controls:

- *Available Plots* list (auto-refresh).
- Buttons:
  - *Open Plot* – External viewer.
  - *Export Plot* – Save in different formats.
  - *Open Results Folder* – File explorer shortcut.
  - *Refresh* – Reloads available plots.
- Viewer Pane: Interactive plot area with zoom/pan, navigation, save, and reset.
- Metadata panel with filename, timestamp, size, and linked simulation ID.

3.9 DRL Tab

Dedicated to training, integration, and evaluation of DRL agents.

Sub-tabs:

1. **Overview**
   - Component availability check (Matplotlib, Pandas, TensorFlow).
   - Workflow instructions.
2. **Training**
   - Select algorithm (PPO/DQN/TDHD), hyperparameters, and number of episodes.
   - Buttons: *Start Training*, *Stop Training*, *View Logs*.
   - Real-time log and graph update.
3. **Results**
   - Displays trained models with name, size, date.
   - Buttons: *Use Default*, *Refresh*, *Open Folder*.
   - Allows model selection for Task Reassignment.

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4. Help System

- The Help button (top right of every tab) loads contextual explanations from rw_fault_help.py.
- Each section corresponds to a tab or sub-tab (overview, fault, ML, DRL, visualization, etc.).
- Help is displayed as a modal window with scrollable text.

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5. Shared Features

- **Run Simulation** – Executes the configured scenario; runs Basilisk, injects faults, triggers ML detection and DRL reassignment, and generates plots and binaries.
- **Auto Save** – Saves configurations before run.
- **Progress Logging** – Each event recorded in simulation log with timestamp.
- **Result Auto-Display** – Switches to "Results" tab on completion.

Part 2 – Instructional User Handbook

Quick Summary of Operation

1. Open the simulator (python core/run_fault_simulator.py).
2. Start in **Constellation** tab to create clusters.
3. Define orbits and satellites.
4. Go to **Fault Configuration** → choose satellites and inject faults.
5. In **Fault Detection**, load ML model and start detection.
6. If DRL is enabled, **Task Reassignment** and **DRL** handle recovery.
7. Use **Targets** and **Visualization** for mission setup and views.
8. Set time in **Output Settings**, click **Run Simulation**.
9. Review results in **Results** tab.
10. Use **Help** anytime for explanations.

Using Each Tab (Step-by-Step)

1. Constellation

- **Create Cluster:** Fill *Cluster Name*, *Number of Satellites*, *Formation Type*, *Separation*, select *Orbit*, then click *Create Cluster*.
- **Clear Form:** Resets all inputs.
- **Existing Clusters:** Use *View Details*, *Modify*, *Delete*, *View Formation*, or *Test Communication*.
- **Orbit Management:** Click “Add Orbit”, set altitude/inclination, save.
- **Satellites:** Use “Add” and “Remove” to adjust constellation.
- **Communication:** View the network map between leader and followers.

2. Fault Configuration

- Select *Cluster* and *Satellite*.
- Check *Enable Fault*.
- Pick a type (Friction, Power, Encoder, Battery).
- Enter Magnitude, Wheel, and Time.
- For recurring issues, enable *Periodic Fault*.
- Click *Apply* to confirm.
- View all configurations under “Fault Summary”.

3. Fault Detection

- Click *Load ML Model* → select anomaly_detection_model.keras.
- Click *Start Detection* to activate analysis.
- Click *Generate XAI* to visualize explanations.
- *Simulate Results* runs test detection without new simulation.
- *View Results* switches to Results tab.

4. Task Reassignment

- Click *Refresh Status* to check DRL system.

- After run, click *View Decisions* to see strategy.
- *Simulate Results* generates synthetic reassignment for testing.
- Explore detailed spacecraft health in "Spacecraft Status".

5. Targets

- *Add Target*: Input name, latitude, longitude, priority.
- *Assign Target*: Choose satellite, click Assign.
- *Auto-Assign All Targets*: Quickly distribute evenly.
- *Check Coverage*: Confirm visible/invisible targets.
- Preview coverage on map.

6. Visualization

- In *Camera & View*, pick a satellite and adjust camera X/Y/Z, FOV.
- Click *Apply Settings*.
- Use *Orbit Control* to show/hide orbit lines.
- Use *Satellite Display* to set color and size.
- Enable *Visualization Preview* to test before running.

7. Output Settings

- Enter simulation time or click *Quick Preset* (10–90 min).
- Check *Show Plots*, *Save Plots*, *Save Binary*.
- Review directories for logs, plots, and Vizfile.
- Change binary name if needed.
- Click *Run Simulation* (top right) to begin.
- Watch log updates below.

8. Results

- Left panel lists plots; click to display.
- Toolbar: zoom, pan, save, reset.
- *Export Plot* → choose format.

- *Open Results Folder* → open containing directory.

9. DRL

- *Training*: Choose PPO/DQN, set episodes, learning rate.
- Click *Start Training*.
- *Results*: Check saved models, use *Refresh* to reload.
- *Use Default* loads standard model for Task Reassignment.

Using Help

- Every tab's *Help* button opens an explanation from rw_fault_help.py.
- You can open help globally via *Help* → *View Tab Help*.
- Help text describes workflow, recommended settings, and error fixes.

Common Tips

- Save configurations before closing.
- For ML/DRL functions, ensure TensorFlow and model files are present.
- The Simulation Log area shows errors in red.
- All results are time-stamped in logs/ and plots/.
- Vizard binaries are stored in Vizfile/_VizFiles/.